STA 334 Statistical Consulting
Fall 2026

Class Meeting Time: MW 2:45-4:00 pm

Location: Fanning Hall 423

Instructor Name and Preferred Title: Professor White

Email: mwhite5@conncoll.edu

Office Hours: MW 4:00 – 5:00 pm and **by appointment.**

Office: TBD

Pre-requisite: STA207 and STA234 or equivalent statistics course. Pre-requisite is MANDATORY. If you have not completed the pre-requisite, talk to me before (preferably) or immediately after the first class.

Course Description: The focus of this course is to have students who:

1. Can communicate statistics to non-statisticians
2. Gain confidence in solving real life problems
3. Are career ready with relevant work experience
4. Are comfortable with a beginner to intermediate level of programming in python.

Software: We will use python software using Jupyter notebooks in Anaconda/Miniconda. You will need a computer with a Mac OS X/ Windows/Linux system. You will also need to bring your laptop to every class.

Moodle: I will share all required information, lecture notes, additional readings, articles, assignments, important announcements, and resources through Moodle. Make sure to check your Moodle account for this course regularly.

Textbook(s): There is no required textbook for the course. All the study and reading material will be shared in class/Moodle.

Class Structure: Class time will consist of lectures, discussions, student presentations and group work. I highly encourage you to participate during the lectures by asking questions, taking part in discussions, making comments, or sharing concerns at any time during the class.

Attention: I would like to emphasize that under any unexpected circumstances, inform me as soon as you can, and I will help you get through the class as much as possible. We will work on this together so ask questions, help each other, and be present.

Class Attendance and Participation: Attendance and activities in the class. The key to consulting is communication, therefore class participation is imperative. Class attendance is *a* necessary or indispensable requirement for effective participation in this class. Please inform me as soon as possible if you will miss a class.

Late Assignment Policy: Start your assignments early so you can get help from me in a timely manner. I will introduce projects and assignments in the classroom to give ample opportunity for questions. In case of unexpected problems, you can talk to me before the due date.

Email: I will utilize your conncoll.edu e-mail addresses & Moodle Announcements. Please use this subject format in communication with me for all matters relating to this class.

STA 334: LastName, FirstName: Topic/Question

Grade: Your final grade will be calculated as the following:

Homework Assignments (25%): Assignments will be based on materials presented or discussed during the class. Assignments with due dates will be shared in class/Moodle.

Final Project (45%): I will share more details in the class. The goal of the final project is to give you an opportunity to showcase the statistical methods you have learned so far in your college career while incorporating aspects of consulting to present information to a non-technical audience.

Class Attendance and Participation (30%): Based on the quality of the student's class participation, the grade determined pursuant to the criteria above may be either increased or decreased to a full grade equivalent. Class participation is *not* performing class assignments or responding when called upon – that is absolutely required by each student. Class participation is the voluntary active engagement in class discussions that reflect intellectual curiosity relating to the subject material. *Notwithstanding the grade achieved pursuant to the percentages above, it is not possible to achieve a course grade in the "A" category (A or A-) without* **effective** *class participation.*

Dean of Faculty Division Communication Process
If you have a concern related to your course, please speak with your instructor directly. If there is an issue that you are not able to adequately resolve, please contact the academic department chair (if your

class is cross-listed, please contact the chair of your instructor's department). If your instructor is the department chair, please contact the Associate Dean of Faculty for Faculty Affairs, Deborah Eastman (daeas@conncoll.edu) or the Associate Dean of Academic Affairs, Chad Jones (cjones8@conncoll.edu) for assistance. You may also consult with your class dean who can provide support to you during this process.

Family Educational Rights and Privacy Act (FERPA)
United States federal law protects access and rights related to educational records. Postsecondary institutional students hold FERPA rights. For further information, visit: https://www.conncoll.edu/academics/registrar/ferpa/.

Office Hours
Office hours provide students with additional opportunities to review or ask questions about the class discussions and assignments. Connecticut College faculty encourage students to go to office hours so they might learn about your interests, both inside and outside the classroom. In addition to talking about class material and assignments, you may find you share common interests, such as music, books, hobbies, and movies. If a professor knows your interest, they may inform you about campus programs and activities or other opportunities like fellowships and scholarships. Most importantly, a professor who knows their students writes better letters of recommendation. Successful students at Connecticut College make time to go to their professors' office hours. All Connecticut College faculty are required to have office hours on their syllabus and posted on their office door. If you cannot make your professor's scheduled office hours, contact your professor to set up an appointment.

Credit Hour Definition
A semester course is normally equivalent to four credit hours. Connecticut College complies with federal regulations defining the credit hour. For each credit hour awarded, a course will provide an average of at least one hour of classroom or direct faculty instruction (class meetings, labs, review sessions, field trips, office hours, film screenings, tutorials, training, rehearsals, etc.) and at least two hours of out-of-class work (homework, preparatory work, practice, rehearsals, etc.) per week.

The Connecticut College Honor Code
Academic integrity is of the utmost importance in maintaining the high standards of scholarship in our community. Academic dishonesty is considered to be a serious offense against the community and represents a significant breach of trust between the professor, the classmates, and the student. There are many forms of academic dishonesty including plagiarism, unauthorized use of AI, falsifying data, misrepresenting class attendance, submitting the same work in two courses without prior approval, unauthorized discussion or distribution of exams or assignments, and offering or receiving unauthorized aid on exams or graded assignments. Students violating the Honor Code may be referred to the college's Honor Council for resolution.

Course Generative AI Policy
You may use AI programs e.g. ChatGPT to help generate ideas and brainstorm only. You cannot submit any work generated by an AI program as your own.

Title IX Statement

As a faculty member, I am deeply invested in the well-being of each student I teach. I am here to assist you with your work in this course. If you come to me with other non-course-related concerns, I will do my best to help. It is important for you to know that all faculty members are trained and required to report any incidents of gender-based discrimination, including discrimination based on gender identity, gender expression, and sexual orientation. This means that I cannot keep information confidential about sex-based harassment, sexual assault, dating violence, domestic violence, stalking, or other forms of gender-based discrimination, and that I will report that information to the Title IX office, if it is shared with me. However, the Title IX office in response to notice from me will reach out to you to offer support and resources, and offer you the opportunity to file a formal Title IX complaint, which is up to you. Nicole Powell, the Director of Sexual Violence Prevention and Advocacy, can advise you confidentially as can Counseling Services and any of the College chaplains or Confidential Advocates. Nicole can also help you access other resources on campus and in the local community. You can reach Nicole at spowell2@conncoll.edu, or make an appointment with Nicole or one of the other Confidential Advocates at https://calendly.com/svpa-confidential-advocates.

The College sexual harassment, dating violence, domestic violence, stalking, and nondiscrimination policies are in the Harassment and Nondiscrimination Policy, Including Sexual Harassment and Nondiscrimination Under Title IX, which can be found here on the College's website. There you will find the policies, definitions, procedures, and resources. If you need to report an incident or have any questions about the policy, you can contact the Title IX Coordinator, Megan D. Monahan, J.D., at 860-439-2624 or mmonahan@conncoll.edu.

Academic Resource Center
The Academic Resource Center (ARC) offers services to support your academic work such as study skills workshops, time management, coaching and tutoring. Its offices are located on the second floor of Shain Library. Students can make appointments by clicking on this link: https://forms.gle/BQecmVdK8Bg1sv5P7.

The ARC is open to the community as a quiet study space at any time the library is open. Professional staff is there for appointments Monday – Friday, 8:30 – 5:00 (evenings are by appointment only). If faculty or students have any questions or concerns, they should contact Patricia Walsh (pdallas@conncoll.edu).

Quantitative Skills Center
The Quantitative Skills Center (QSC) provides and facilitates free services and resources to support students engaged in quantitative work across the curriculum. Quantitative skills encompass the proficiency in using mathematical and statistical tools in a variety of contexts. The center offers group and drop-in sessions as well as peer-to-peer tutoring. The tutoring services aim to foster self-directed and reflective learning, empowering students to employ effective strategies for academic success. Please fill out this form to schedule an appointment. For more information, please email skilicbah@conncoll.edu or pdallas@conncoll.edu.

Writing Center
The Roth Writing Center provides one-to-one peer tutoring (free of charge) to help student writers of all abilities during all stages of the writing process. If you're a confident, experienced writer, our tutors can help you to push your ideas and polish your style; if you're a relatively inexperienced and not-so-confident writer, they can help you to work on grammar or organization or whatever you

need. Working with a tutor gives you the opportunity to share your work-in-progress with an actual reader so that you can get useful feedback on that work *before* you have to turn it in for a final grade. You can make an appointment by using the Google Calendar link on the Writing Center's website or by emailing the Writing Center at writingcenter@conncoll.edu; a new calendar of appointments will become available by the second week of each semester.

Office of Student Accessibility Services
Connecticut College complies with Section 504 of the Rehabilitation Act of 1973 and the Americans with Disabilities Act. If you have a documented disability and have been approved for academic accommodations, please access your AIM Student Portal and request to have your Faculty Notification Letters emailed to your professors at the start of the semester. If you are not approved for accommodations, but have a disability requiring academic accommodations, or have questions about applying for accommodations, please contact Student Accessibility Services at 860-439-5428 or sas@conncoll.edu.

Shain Library Research Assistance
Whether you are starting to brainstorm ideas for a paper, or need help finding articles, books, data, and other media, or if you just need help managing citations, the research support librarians in Shain Library are here to help you navigate the world of libraries and information at every stage of the research process. We offer individualized research consultations during normal business hours Monday through Friday. To ask a question, start a conversation, or schedule a research consultation, please get in touch with us here: https://conncoll.libanswers.com/

Classroom Recording
With the exception of those granted accommodations through the Office of Student Accessibility Services, students are prohibited from audio, video, or photographic recording during class periods or out-of-class meetings with the instructor without explicit permission from the instructor. Recordings approved in this manner may not be shared in any form without permission of the instructor. Violations of this policy shall be considered an Honor Code violation.

Respecting Personal Pronouns and Identity
Everyone deserves to be referred to and addressed in accordance with their personal identity. As a faculty member, I am committed to ensuring my classroom affirms people of all gender expressions and gender identities. In this course, we will only use the name and pronouns of each individual's choosing. The repeated usage of incorrect names and/or pronouns may constitute a violation of Connecticut College policy as well as a violation of law. If you have any specific questions regarding this policy, email the Title IX Coordinator, Megan Monahan at mmonahan@conncoll.edu.

In the classroom, be assured that you will always be referred to by the name and pronouns you choose. If you go by a different name than your legal name, Connecticut College has a process to change your preferred name on most campus systems. If you want to learn more about this process go to conncoll.edu/equity-inclusion/preferred-name-faq/ or email GSP@conncoll.edu.

Students, faculty and staff are now able to choose and share their pronouns within the college community by using the Preferred Name/Pronouns link on the Account navigation menu in CamelWeb and the CC Mobile App. Your gender pronouns will appear in the internal directory located in CamelWeb and the CC Mobile App. If none are selected, or if "Not Applicable" is

selected, no pronouns will display. Enrolled students' gender pronouns will also display in Moodle for instructors via the class participants page.

Pronouns are one way to affirm someone's gender identity, but they are not necessarily indicative of a person's gender identity. Commonly, they/them is a gender-inclusive pronoun used by a variety of identities. However, while some people use they/them, others may use pronouns like ze/zem, xi/xim, he/him, she/her, any combination of those and/or many others. They may even reject pronouns altogether and use their name in place of pronouns. Remember to ask for pronouns, listen, and then respect the gender identities of those around you by using the proper terminology. If you have any further questions, please do not hesitate to contact Nicole Powell at spowell2@conncoll.edu.

Tentative Plan for STA-334

Week	Date	Topic	Homework Assignment
1	Sept 2 (W)	What is Statistical Consulting?	
2	Sept 7 (M)	Course Overview, Syllabus, Introduction to Statistical Consulting	
2	Sept 9 (W)	Software Installation and Troubleshooting; Introduction to Python and Jupyter Notebooks	
3	Sept 14 (M)	Introduction to Python Continued Introduction to AE and SV Datasets	HW 1 Assigned; Due Sept 25
3	Sept 16 (W)	Final Project Introduction and Deliverables	HW 2 Assigned (Final Project Dataset Selection and EDA Notebook); Due Oct 2
4	Sept 21 (M)	**NO CLASS (YOM KIPPUR)**	
4	Sept 23 (W)	Final Project Questions and Discussion	HW 1 Due Sept 25 Final Project proposal, progress and final deliverables assigned
5	Sept 28 (M)	Mock Consultation 1	
5	Sept 30 (W)	HW 1 Review	

6	Oct 5 (M)	Statistics Packages in Python; Visualization in Python	HW 2 Due Oct 2
6	Oct 7 (W)	Mock Consultation 2	
7	Oct 12 (M)	**Self-Directed Learning**	
7	Oct 14 (W)	**Self-Directed Learning**	Final Project Proposal Due Oct 16
8	Oct 19 (M)	**FALL BREAK**	
8	Oct 21 (W)	**Self-Directed Learning**	
9	Oct 26 (M)	Final Project Questions and Discussion	
9	Oct 28 (W)	Mock Consultation 3	
10	Nov 2 (M)	Class Discussion - Translating statistical findings into client-friendly language.	**Course Withdrawal Period Ends**
10	Nov 4 (W)	Class Discussion - Translating statistical findings into client-friendly language.	Final Project Progress Report Due Nov 5
11	Nov 9 (M)	Final Presentations and deliverables- Questions and Discussion	
11	Nov 11 (W)	Final Project Presentations	
12	Nov 16 (M)	Final Project Presentations	
12	Nov 18 (W)	Final Project Presentations	
13	Nov 23 (M)	**THANKSGIVING BREAK**	
13	Nov 25 (W)	**THANKSGIVING BREAK**	
14	Nov 30 (M)	Final Project Presentations	
14	Dec 2 (W)	Final Project Presentations	
15	Dec 7 (M)	Final Project Presentations	

15	Dec 9 (W)	Course Wrap-Up, Final Project Presentations (if needed)	**Last Class Meeting**
Final	Dec 16		**Final Project Deliverables Due**

Final Project

45% Final Project: Each student will pick a consulting topic, prepare a writeup and make a class presentation. The presentation should describe the problem considered, why it is of interest, the data available, and the goals of inference. Then the method of data analysis should be discussed, the results of that analysis, along with the conclusions made and a sense of how reliable those conclusions are.

Projects

As you begin thinking of a potential project, please keep the following items in mind:

1. The overall question or questions you would like to address.

2. What data you will use and where it can be obtained.

3. What specific predictors are available, roughly how many there are, and how large a sample size you will have.

4. What specific response you would like to predict, and what model and methods you will use to predict it.

Data Links of Interest

- CDC On Line Data Bases
- www.data.gov
- http://sda.berkeley.edu/
- http://archive.ics.uci.edu/ml/datasets.html
- http://www.bigdata-startups.com/public-data/
- http://www.statsci.org/datasets.html
- http://www.pro-football-reference.com/play-index/play_finder.cgi
- https://www.kaggle.com/
- https://www.rita.dot.gov/bts/data_and_statistics/index.html
- http://investexcel.net/multiple-stock-quote-downloader-for-excel/

Final Project Deliverables (45% of grade)

Proposal Document (5%) Final Project Proposal Document Due 11:59PM October 16th

1. Introduction
2. Methodology
3. Project Plan
4. References

Progress Report (10%) Final Project Progress Report Due 11:59PM November 5th

1. Introduction
2. Methodology and Results
3. Project Plan
4. References

Final Report (30%) Due 11:59PM December 16th

1. Code (in Python or R) (**2.5%)**
2. README.txt file – What I need to install so I can run your code (**2.5%)**

3. Presentation slides and in-class presentation (20**%)**
4. Review of peer presentations – **(5%)**

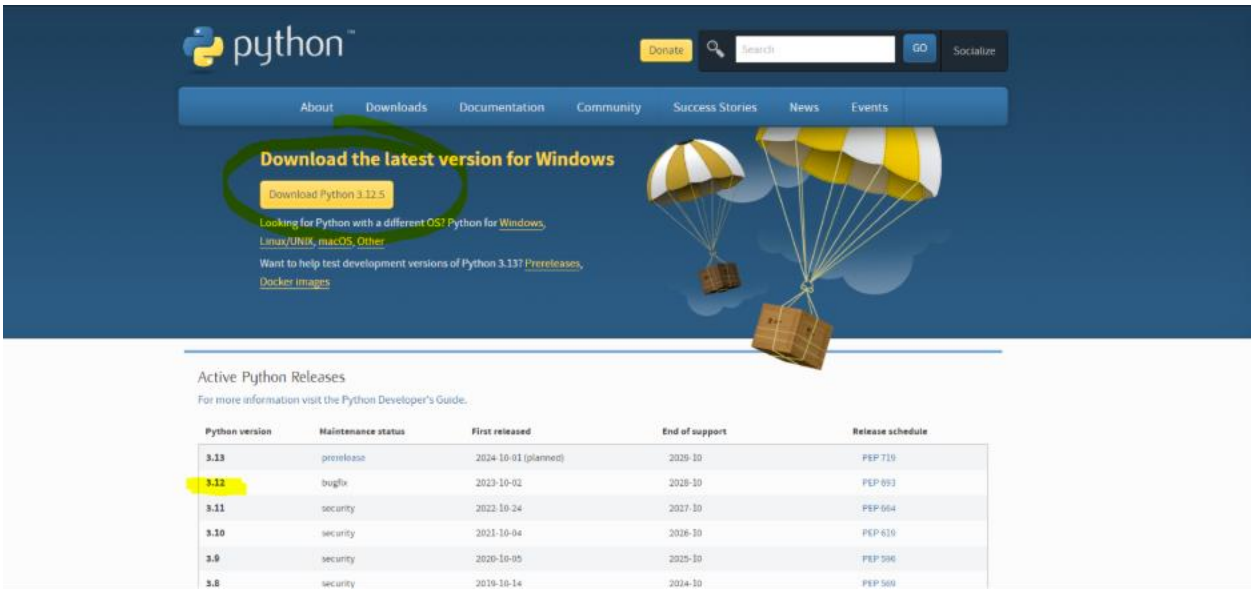
Final Deliverables Grading Rubric

Proposal and Progress Documents	1. Introduction 2. Methodology 3. Project Plan 4. References Document contains all above sections with extensive elaboration, details in ALL sections and visualizations included.	1. Introduction 2. Methodology 3. Project Plan 4. References Document does not contain all above sections (-5 Points for each missing section) Elaboration, details limited in one or more sections. Few visualizations included (-2 points for less elaboration, limited details)	Assignment not turned in.

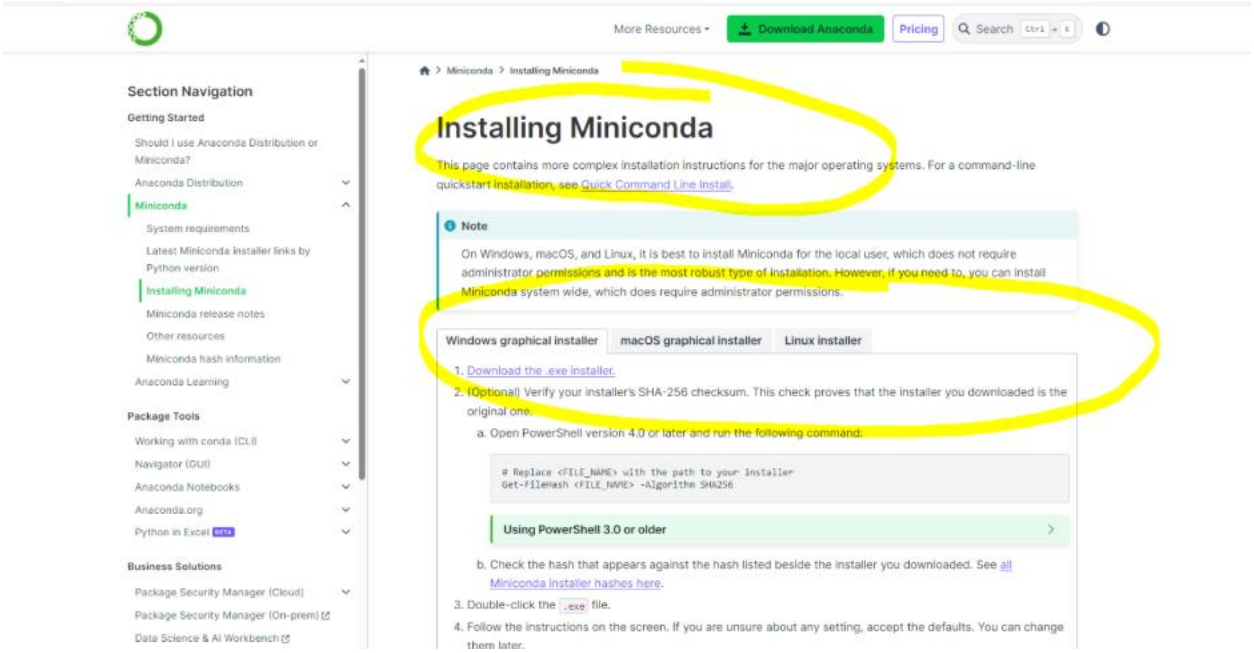
	100 points	-2 points if NO visualizations included	
Presentation	1. Clear business problem. 2. Background / nature of the business problem. 3. Data used is described in detail. 4. Visualizations are included 5. Methods used in the analysis using non statistical language 6. Issues in/with the analysis 7. Conclusion using non statistical language All above are present There is clear elaboration on all the above 100 points	1. Clear business problem. 2. Background / nature of the business problem. 3. Data used is described in detail. 4. Visualizations are included 5. Methods used in the analysis using non statistical language 6. Issues in/with the analysis 7. Conclusion using non statistical language Any of the above are missing (minus 10 points) Elaboration is missing or weak on one or more the above. (minus 5 points)	Does not present 0 points

Python Environment setup

Python - https://www.python.org/downloads/



Miniconda or Anaconda install - https://docs.anaconda.com/miniconda/



Miniconda vs. Anaconda - https://docs.conda.io/projects/conda/en/stable/user-guide/install/index.html